Every plane section of a sphere is a circle. The line of section of a sphere by a plane passing through the centre of the sphere is called a great circle all other plane sections are called small circles.

As shown in the diagram, circle with CD as diameter is a great circle, whereas, the circle with QR as diameter is a small circle.

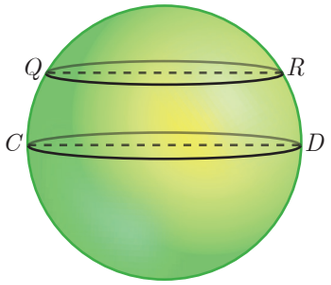


Fig. 7.14

Surface area of a sphere

Archimedes Proof

Place a sphere inside a right circular cylinder of equal diameter and height. Then the height of the cylinder will be the diameter of the sphere. In this case, Archimedes proved that the outer area of the sphere is same as curved surface area of the cylinder.

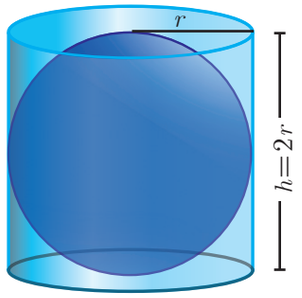


Fig. 7.15

Surface area of sphere =curved surface area of cylinder

$$= 2\pi rh = 2\pi r\left(2r\right)$$

Surface area of a sphere $= 4\pi r^2$ sq.units

Activity 2

(i) Take a sphere of radius 'r'.

(ii) Take a cylinder whose base diameter and height are equal to the diameter of the sphere.

(iii) Now, roll thread around the surface of the sphere and the cylinder without overlapping and leaving space between the threads.

(iv) Now compare the length of the two threads in both the cases.

(v) Use this information to find surface area of sphere.

7.2.5 Hemisphere

A section of the sphere cut by a plane through any of its great circle is a hemisphere.

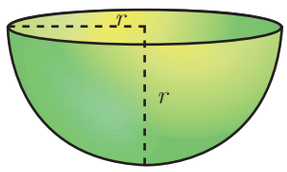


Fig. 7.16

By doing this, we observe that a hemisphere is exactly half the portion of the sphere.

$$\text{Curved surface area of hemisphere} = \frac{\text{C.S.A. of the sphere}}{2} = \frac{4\pi r^2}{2}$$

C.S.A. of a hemisphere $= 2\pi r^2$ sq.units

$$\text{Total surface area of hemisphere} = \text{C.S.A.} + \text{Area of top circular region}$$
$$= 2\pi r^2 + \pi r^2$$

T.S.A. of a hemisphere $= 3\pi r^2$ sq.units

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